



Rachamalla Prashanth

Roll No.: 230102075

B.Tech - Electronics and Communication Engineering

Minor in Computer Science and Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.71 (Current)	2023-Present
B.Tech. Minor	Indian Institute of Technology, Guwahati	7.5 (Current)	2024-Present
Senior Secondary	State Board of Intermediate Education, Telangana	98.4%	2023
Secondary	Board of Secondary Education, Andhra Pradesh	100%	2021

EXPERIENCE

- Tech Mahindra** May 2026 - Jul. 2026
Software Engineer Intern Hyderabad, India
 - Built a **production-grade Node.js backend automation ecosystem** eliminating manual L1.5 ticket handling for a **Direct Client requirement**, using decoupled **REST APIs** to orchestrate IAM workflows across **3 enterprise platforms**.
 - Architected a **real-time event-driven pipeline** with **PostgreSQL LISTEN/NOTIFY** and **SSE**, powering a stateful multi-step orchestrator with **LLM-powered NLP** for routing unstructured support tickets and a **7 table normalised schema**.
 - Automated **4 IAM workflows end-to-end**, replacing repetitive multi-window operations involving **7-8 L1.5 support personnel** with a **single admin click**, significantly reducing manual effort and operational turnaround.

PROJECTS

- DocuMind- Intelligent Document Assistant (RAG)** May 2025 - Jul. 2025
Group project GitHub
 - Engineered an intelligent document Q&A system with a GPU-accelerated RAG using PyTorch CUDA, Faiss, FastAPI, PostgreSQL and React, achieving embedding generation on GPUs with automatic CPU fallback.
 - Built a multi-format document pipeline for PDF, DOCX, TXT, Markdown, and HTML using PyPDF2, python-docx, and BeautifulSoup, integrating LLMs and persistent FAISS indexing.
 - Built an analytics dashboard in ReactJS visualizing search latency and GPU vs CPU throughput, with memory optimizations.
- Image Captioner** Nov. 2024 - Feb. 2025
Group Project GitHub | Website
 - Designed an encoder-decoder image captioning pipeline using CNN for feature extraction and LSTM for sequence generation.
 - Built a preprocessing module for image feature extraction, caption cleaning, tokenization, and sequence creation to prepare image-text data for training. Implemented both greedy search and beam search decoding algorithms to generate captions.
 - Implemented greedy and beam search decoding for caption generation and evaluated model quality using BLEU-1 to BLEU-4.
- VaaniSeva — Multilingual Government Services Assistant** Jul. 2026 - Aug. 2026
Personal project GitHub
 - Engineered a government services assistant that combines deterministic intent handling for common service requests with an **LLM fallback for open ended queries**, providing reliable responses without depending on LLM for every interaction.
 - Enabled **language-aware interactions** across English, Telugu, Hindi, and Kannada using lightweight **Unicode-based detection**, allowing citizens to communicate naturally in their preferred language while maintaining consistent service interactions.
 - Built a modular backend with **FastAPI** with automated unit tests covering multilingual intent detection and AI fallback.

TECHNICAL SKILLS

- Programming:** C, C++, JavaScript, Python
 - Operating Systems:** Windows, Linux*
 - Database Management:** MySQL, PostgreSQL, MongoDB*
 - Web Technologies:** HTML, CSS, ReactJS, NodeJS, ExpressJS
 - Miscellaneous:** Git, GitHub, LTspice, MATLAB*, Verilog*
- * Elementary proficiency

KEY COURSES TAKEN

- Computer Science:** Data Structures and Algorithms, Operating Systems, Database Systems, OOPS and Data Structures, Digital Logic and Computer Architecture, Computer System Design, Machine Intelligence
- Mathematics:** Linear Algebra, Probability and Random Processes, Basic Calculus, Real and Complex Analysis
- Miscellaneous:** Embedded Systems, IT and Coding, Introduction to Mathematical Economics, Industrial Organisation

ACHIEVEMENTS

- Codeforces:** Achieved a global rank of **1868** in Educational Round 189 (Div. 2) among **32000+** candidates. Handle
- CodeChef:** Achieved a global rank of **1362** in Starters 208 among **29000+** candidates. Handle
- NxtWave launchpad:** Achieved a global rank of **3** in Nxtwave Weekly Contest 13 (Advanced).
- Flipkart Grid 8.0:** National **Semi-Finalist** among **1,65,000+** participants.
- AP EAPCET 2023:** Achieved a rank of **486** among **2.3 lakh** students.
- TS EAMCET 2023:** Achieved a rank of **639** among **1.95 lakh** students.

EXTRACURRICULARS

- WebOps Executive, TEDxIITGuwahati:** Worked as a WebOps team member to build and maintain the website. The website is now live and actively used and attracted **1000+** unique users to the website.
- City Representative, Anantapur, Technothon:** Managed to bring a participation of **100+** students from various schools in the city and conducted the exam smoothly.
- Organiser, Technothon:** Organised a **Finance** based event called Fauna Fantastico for the Top **50** Finalists.
- NSS Volunteer:** Engaged in various community service activities, including health camps and cleanliness drives through IIT Guwahati, witnessing a significant footfall of **1000+** students.
- Mentorship, Cepstrum:** Mentoring 2 second-year EEE students, fostering their academic, professional growth.
- Inter Hostel Sports Meet, Spardha 2024:** Represented Brahmaputra Hostel in the inter-hostel parade competition among **13** hostels, demonstrating team spirit, discipline, and active participation in hostel-level events.